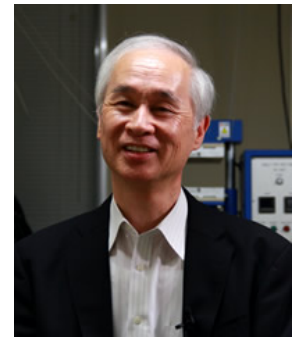


Introduction of High efficiency Rare Elements Extraction Technology in Tohoku University

Atsushi Iizuka, Etsuro Shibata,
Takashi Nakamura

Institute of Multidisciplinary Research for Advanced
Materials, Tohoku University



Project Leader
Prof. Takashi Nakamura

Collaboration of ministries for recycling and circulation of critical elements from Urban Mining

MOE

Ministry of the Environment

Recycling society system

- System building of recycling of small domestic electric appliances
- Promotion of R&D of recovering and waste treatment of critical elements from waste

practical application

Needs from social system

MEXT

Ministry of Education, Culture, Sports, Science and Technology

R&D of technology for "Urban Mining"

- Fundamental science of recycling & testing of critical elements
- Scientific investigation on the contribution of circulation of critical elements to low carbon society

Practical application

Needs from industries

METI

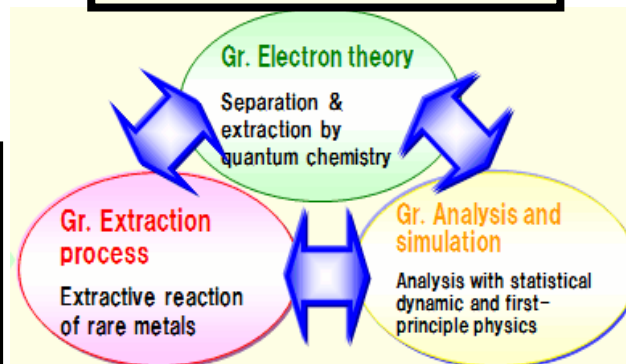
Ministry of Economy, Trade and Industry

Promotion of 3R(Reduce, Reuse, Recycle)

- Recycling technology of REE and critical elements
- Recovering technology of critical elements from end-of-life products; magnet, fluorescent, polisher etc.
- Critical elements recycling technology from process waste

Expected results of collab.

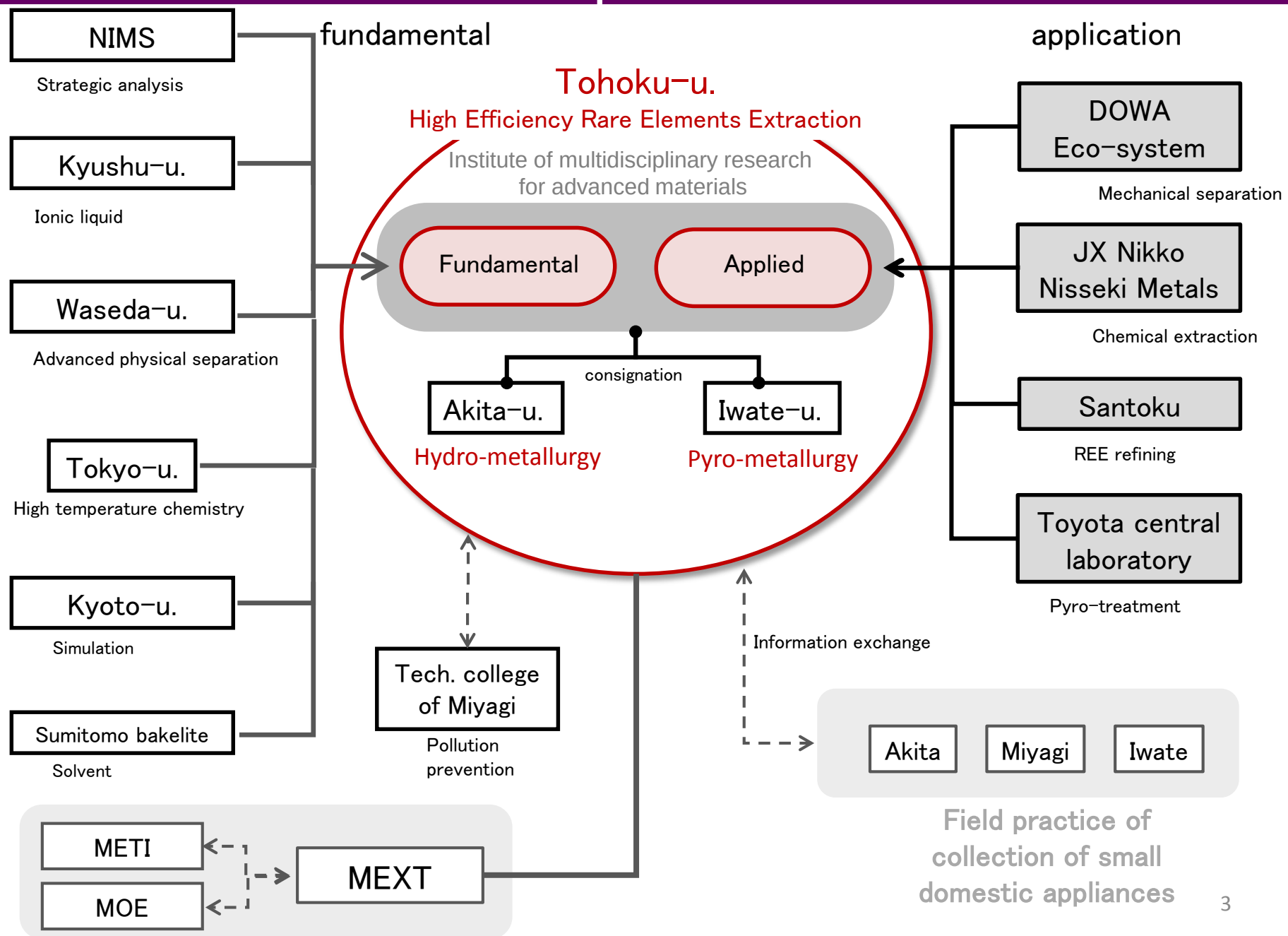
- Properly disposal & recycling of hazardous waste
- Nondestructive analysis of critical elements from waste



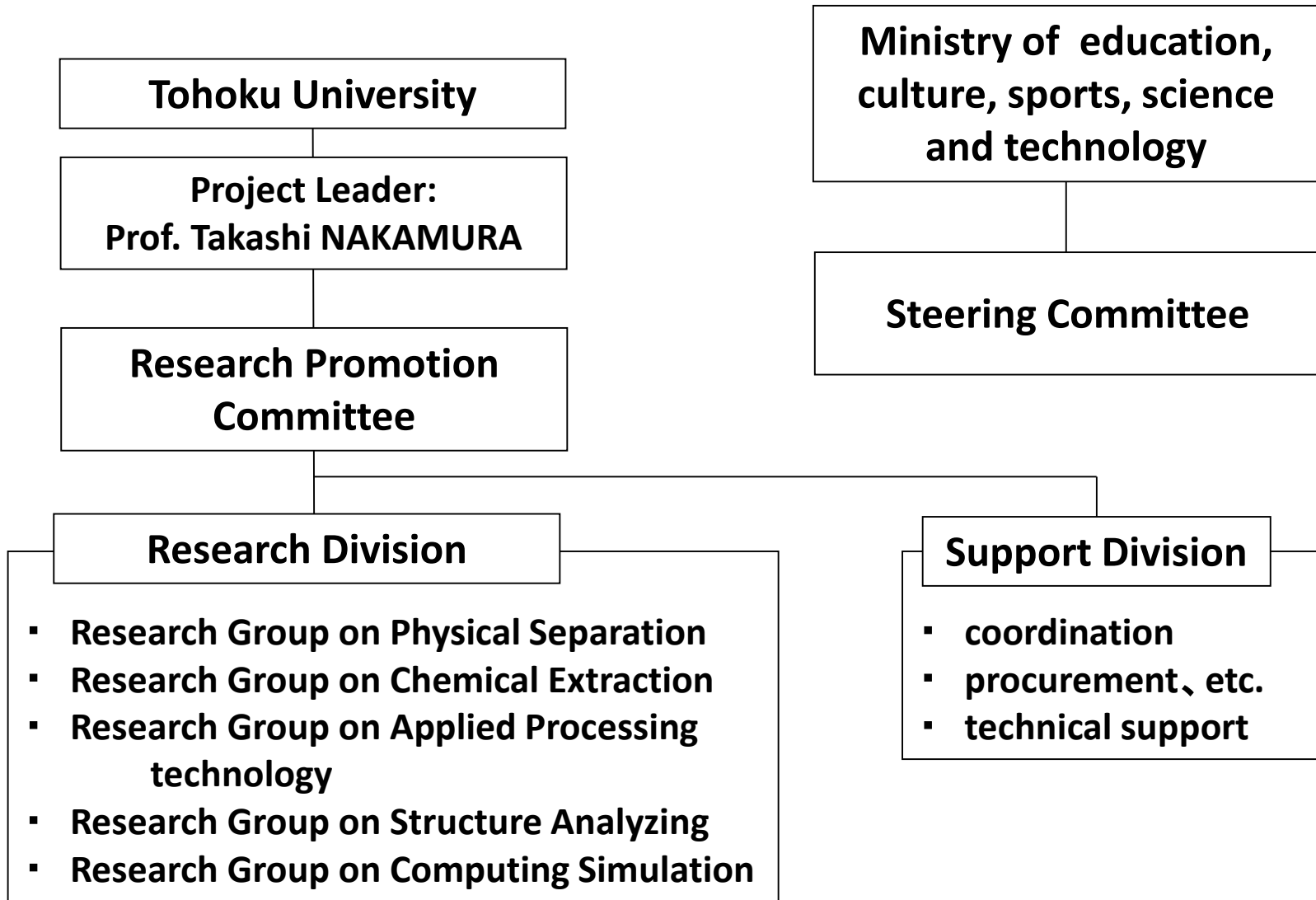
Expected results of collab.

- High efficient extraction process with ionic state
- Low cost automatic beneficiation for recycling²

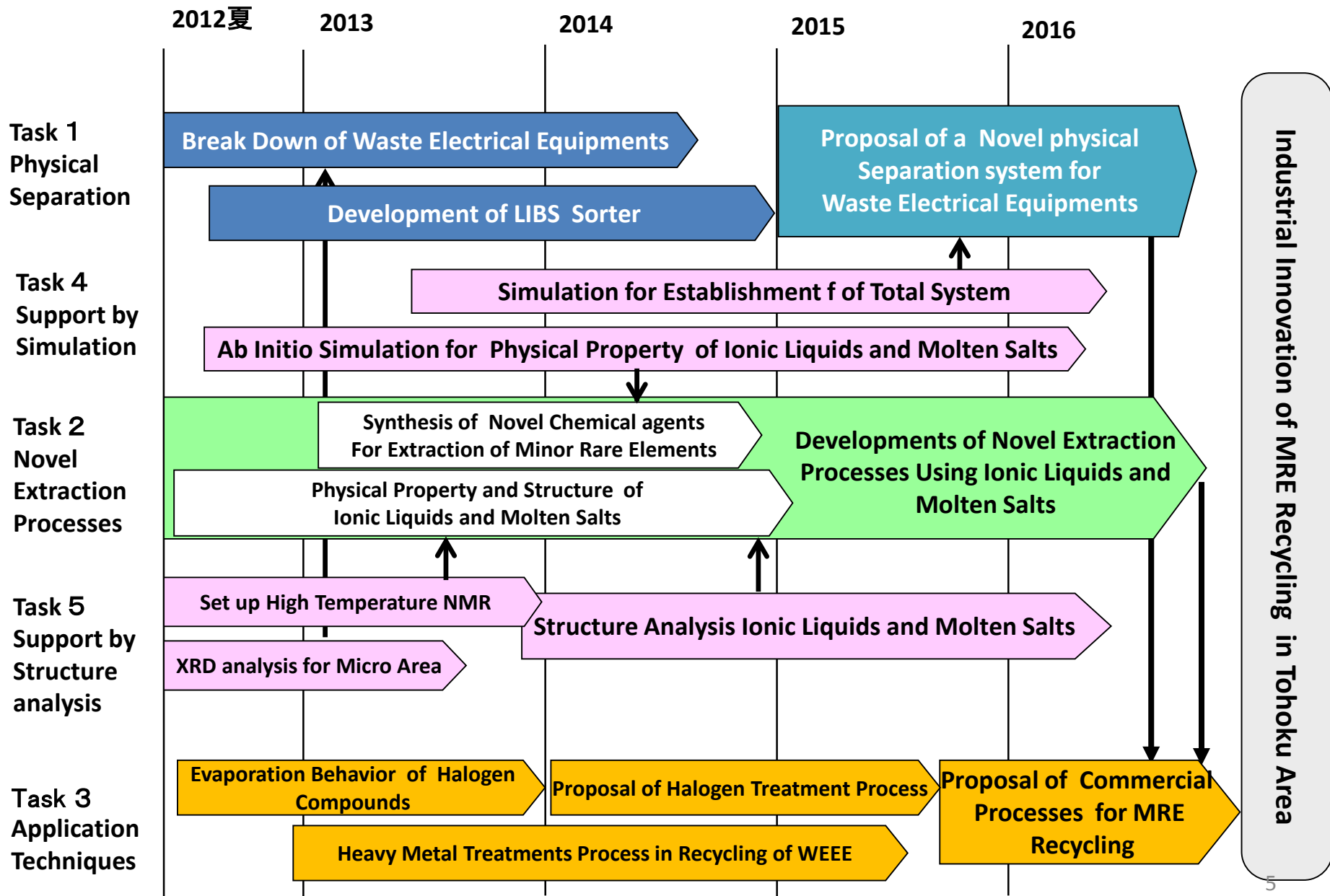
TIMT - High Efficiency Rare Elements Extraction Structure for cooperation



Organization Chart



Research Roadmap



Physical Separation for WEEE

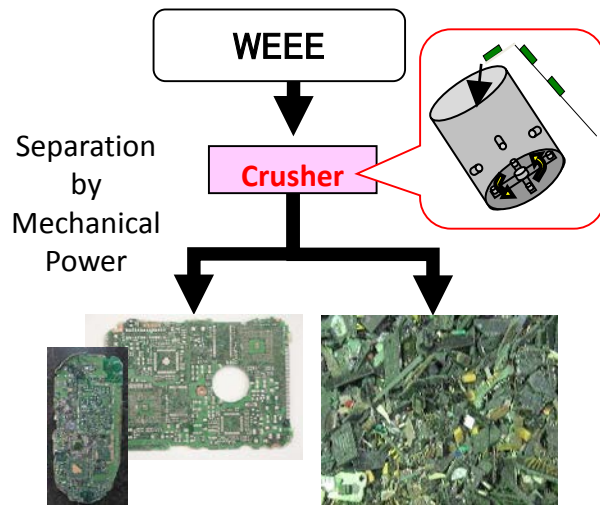
Effective Recovery of Devices
containing Minor Rare Elements
from WEEE

Developments of Novel Break Down Technology

Devise Separation from WEEE

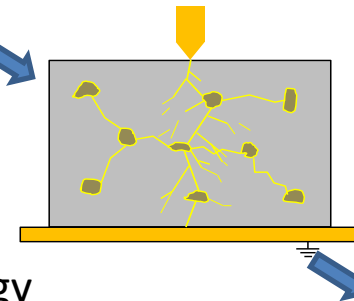
• Conventional Crushers and Mills are strong comminution methods and then destroy the devices in WEEE. They give too small size particles which containing minor rare elements . It is not effective to recycling minor rare elements .

⇒ **Novel Break Down Method for Waste Electrical Equipments is necessary**
Electrical Disintegration method using High Voltage Electric Pulse
is one of candidates



Electrical Disintegration Method by Electric Pulse

New Technology



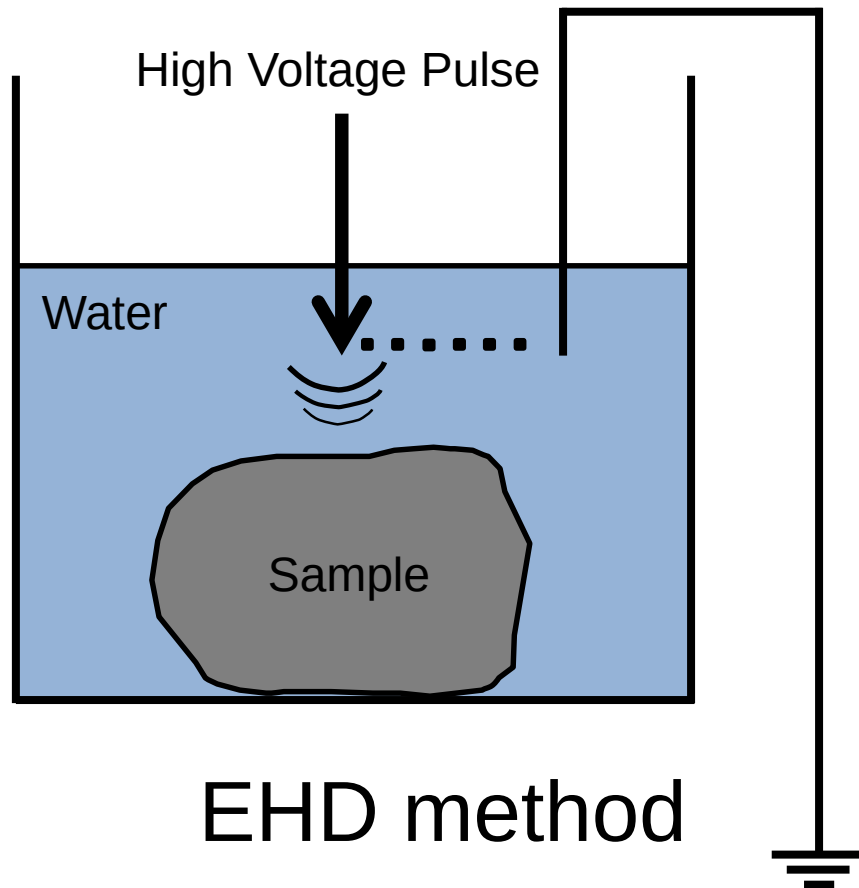
Possible Separation
of devises from WEEE



Devises⁷

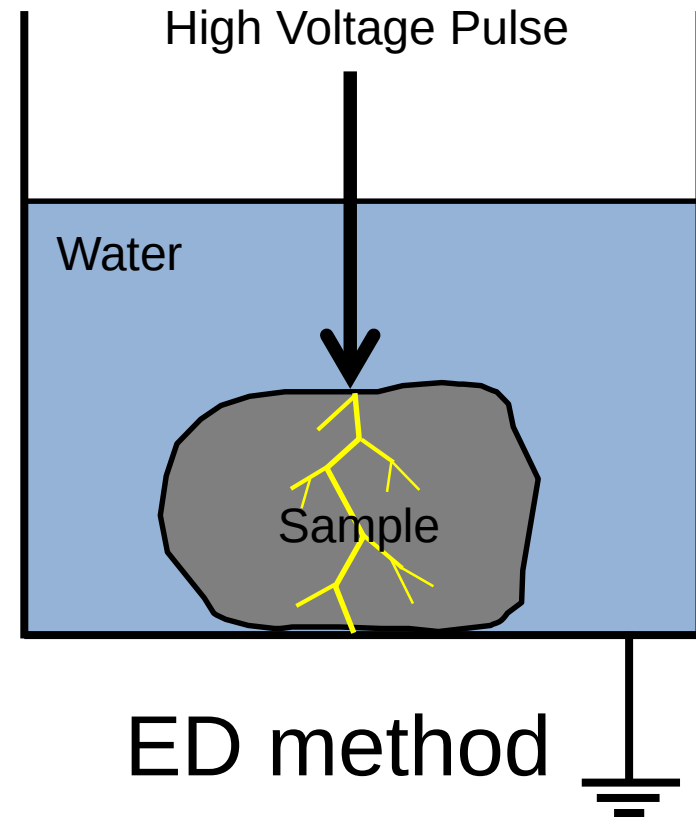
Difficult to separation of devises from WEEE

Electrical Disintegration Method by Electric Pulse



ElectroHydraulic Disintegration

Comminution of WEEE by Shock Wave
In Water



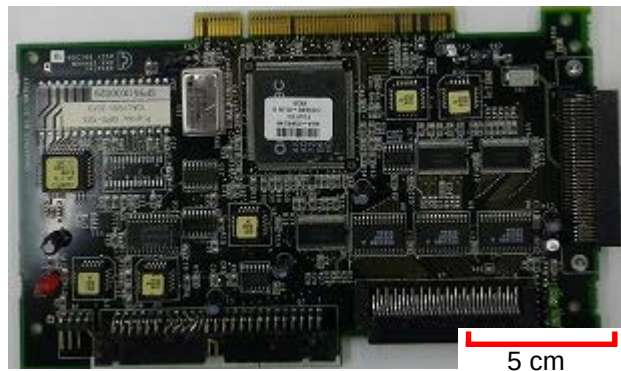
Electrical Disintegration

Comminution of WEEE by Direct
Electric Pulse in Solid

Samples and Apparatus

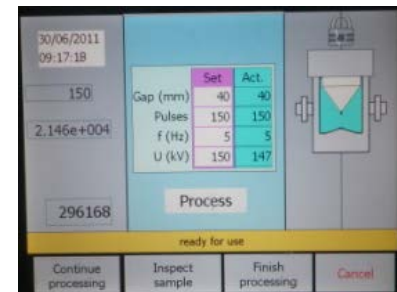
Sample

16 of Print Circuit Boards
Total weight 11.3 kg



Apparatus

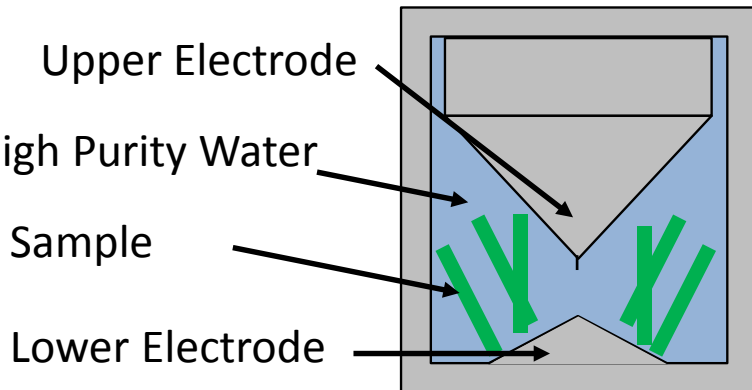
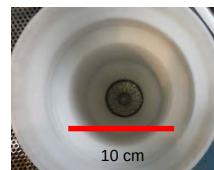
Pulverizer by Electric Pulse developed by SELFRAG AG



Control Panel



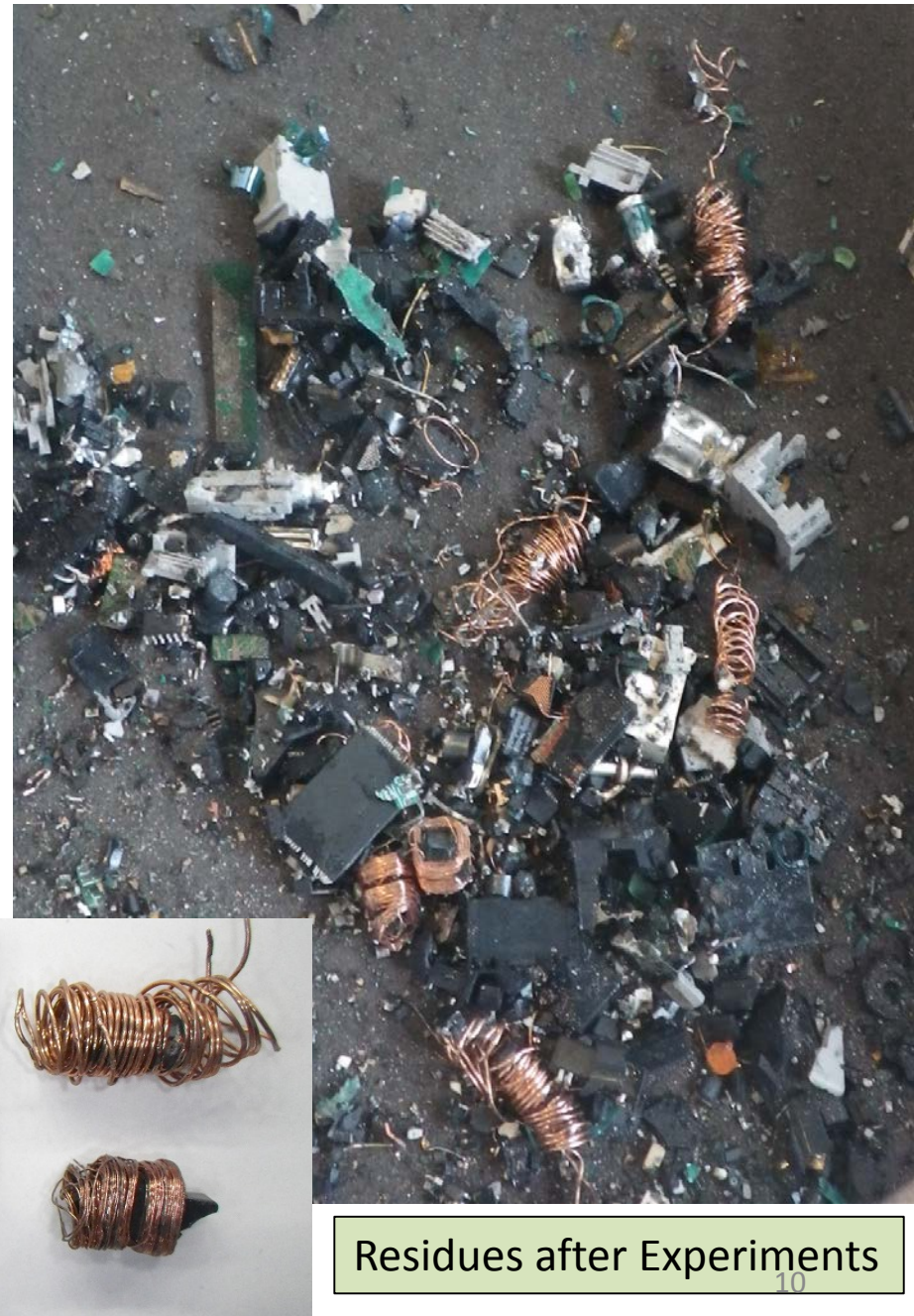
Upper Electrode
Vessel



Example 1



PCB before Experiments

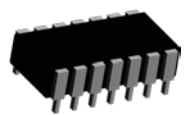
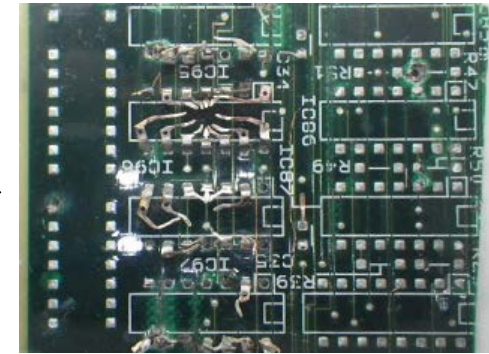
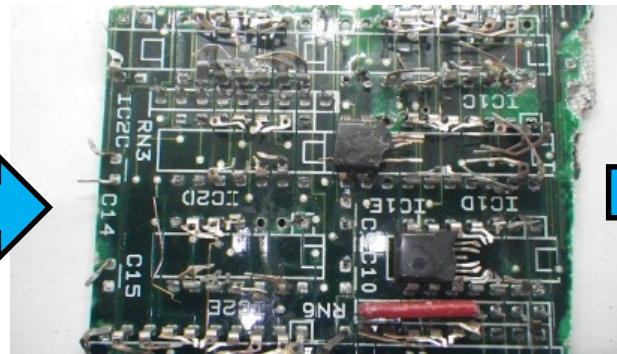
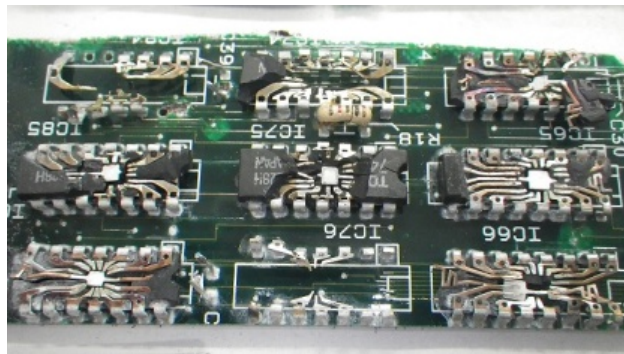


Residues after Experiments

Example 2

Check differences of samples

Devices using lead lines



IC chips

electric pulse passed between plastics and metals



Metal parts inside



Metal leads



Plastics

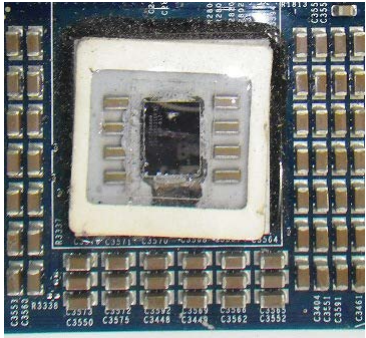


Another metal parts

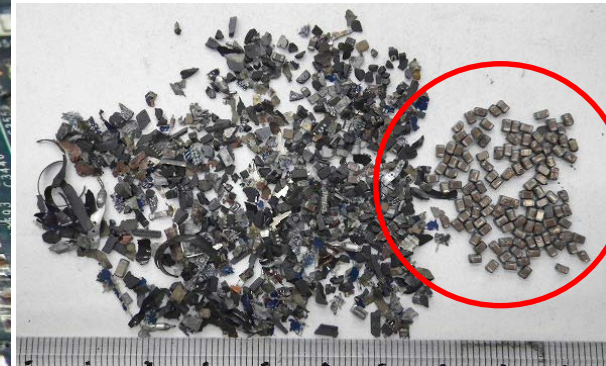
Example 3

Check differences of samples

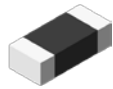
Small parts on PCB



After Pulverization



recovery in
Non- destroyed shape



Ceramic condensers
(1~2 mm)



Ceramic condensers recovered



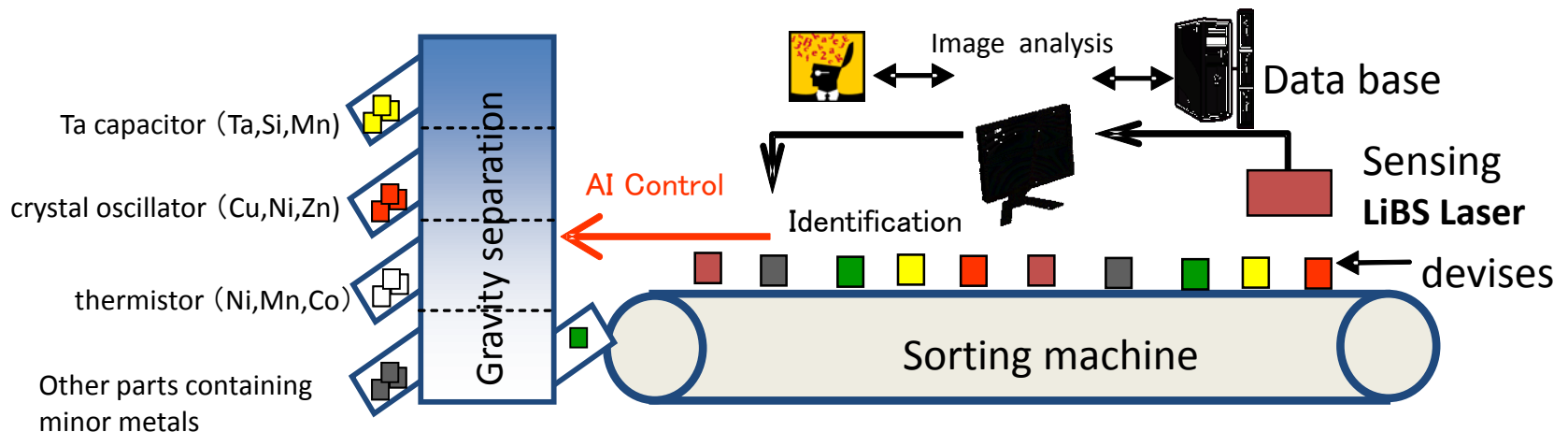
Small parts detached after Pulverization

From Prof.Owada , Waseda University

Automatic Separation of devices and parts containing minor metals from WEEE

content

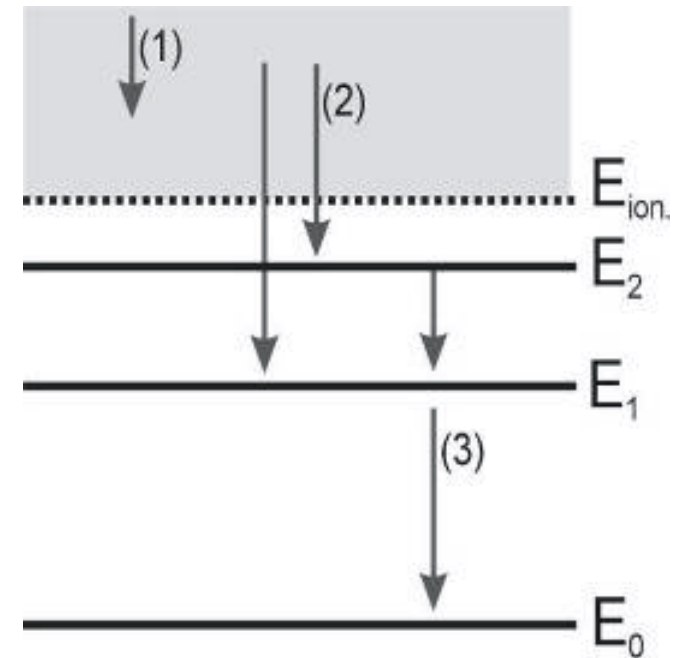
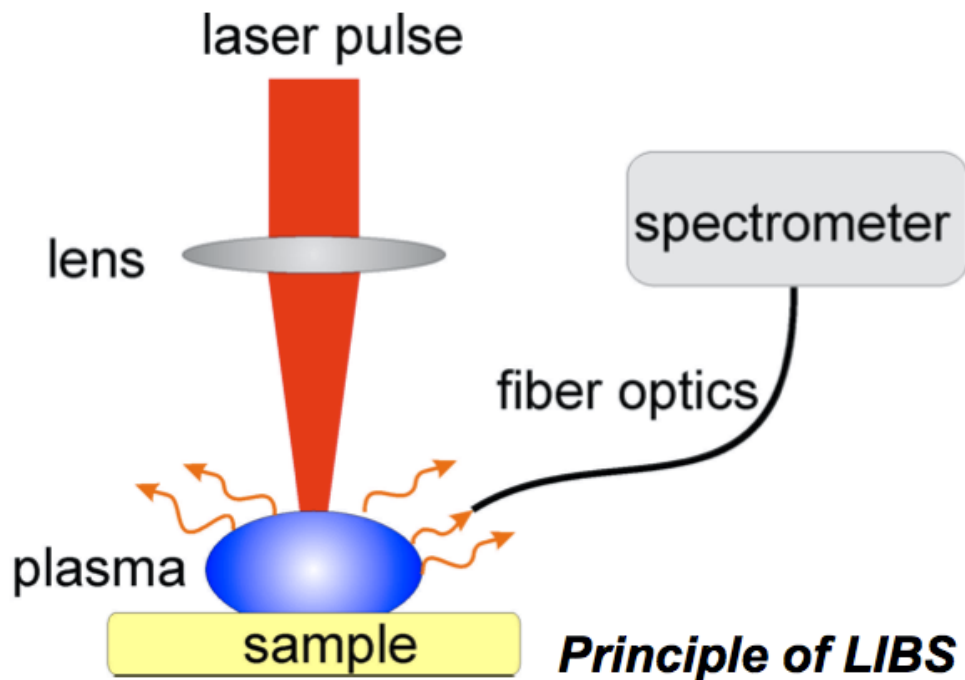
Developments automatic system which combines gravity separation and sorting separation using IT techniques.



- Development of gravity separator which can separate each device
- Sorting machine which can separate each device

Key points: Smart system consist of each separation technique

Laser-induced breakdown spectroscopy (LIBS)



All elements, even light atoms, are analyzed in order of ppm and sample size is enough also within mm order
We are planning to build the system in the next year.

Chemical Extraction of Minor Rare Metals

Developments of High efficient and
High quality separation of Minor Rare
Metals in Ionic Liquids and Molten Salts

Research subjects on Chemical Extraction of Minor Rare Metals

(1) Ionic Liquids

- **Synthesis novel ionic liquids for electro-winning of MRE.**
- **High efficient extraction agents for the ionic liquids.**
- **Physical Properties measurements and simulation by first principal method**
- **Effect of H₂O in Ionic Liquids**

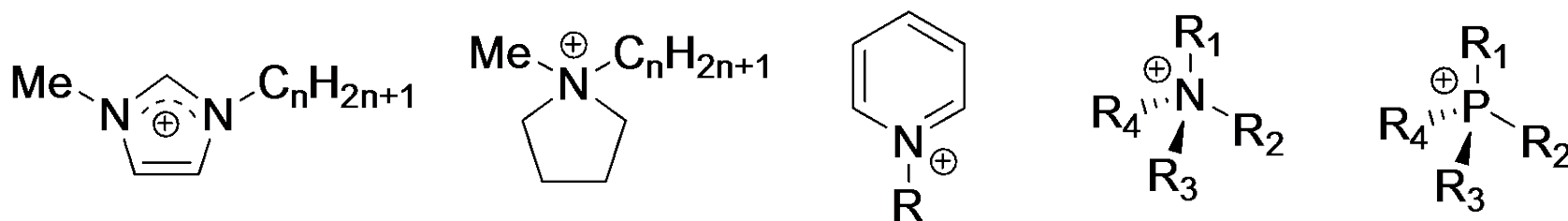
(2) Molten Salts

- **High efficiency electro-winning of Rare Earth Elements in new complex salts**
- **Clarify relationship between physical properties and structures**

(3) New extraction agents in solvent extraction, especially for PGM and Rare Earth Elements(for example Nd from Pr)

Ionic Liquids for Extraction of MRE

Typical cations



Typical anions

NTf_2

PF_6

BF_4

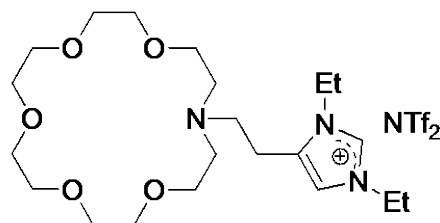
CF_3SO_3

Cl

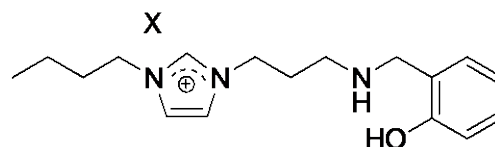
Br

ClO_4

Task specific ionic liquids

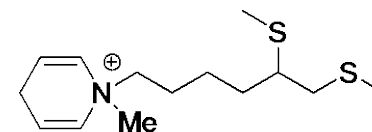


$\text{Cs}^+, \text{Sr}^{2+}$



$\text{X} = \text{NTf}_2, \text{PF}_6$

Am^{3+}



$\text{Zn}^{2+}, \text{Cd}^{2+}, \text{Hg}^{2+}, \text{Tl}^+, \text{Pb}^{2+}, \text{Sn}^{4+}, \text{Ba}^{2+}$

COSMO-RS Method for Simulation for Estimation of Physical Property

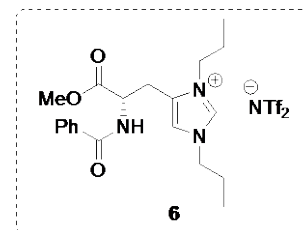
Input : Molecular structure of interest, X



Quantum chemical COSMO calculation

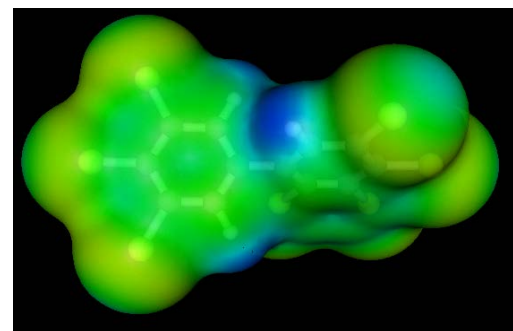
First step ;

DFT / COSMO (Gaussian)



Output :

- Geometry of molecules
- Total energy of molecules in its conductor, E_{COSMO}
- 3D information about polarization charge density on their surface, σ



Blue color : positive charge



• σ - profiles of the molecule, $p^x(\sigma)$



Statistical Thermodynamics



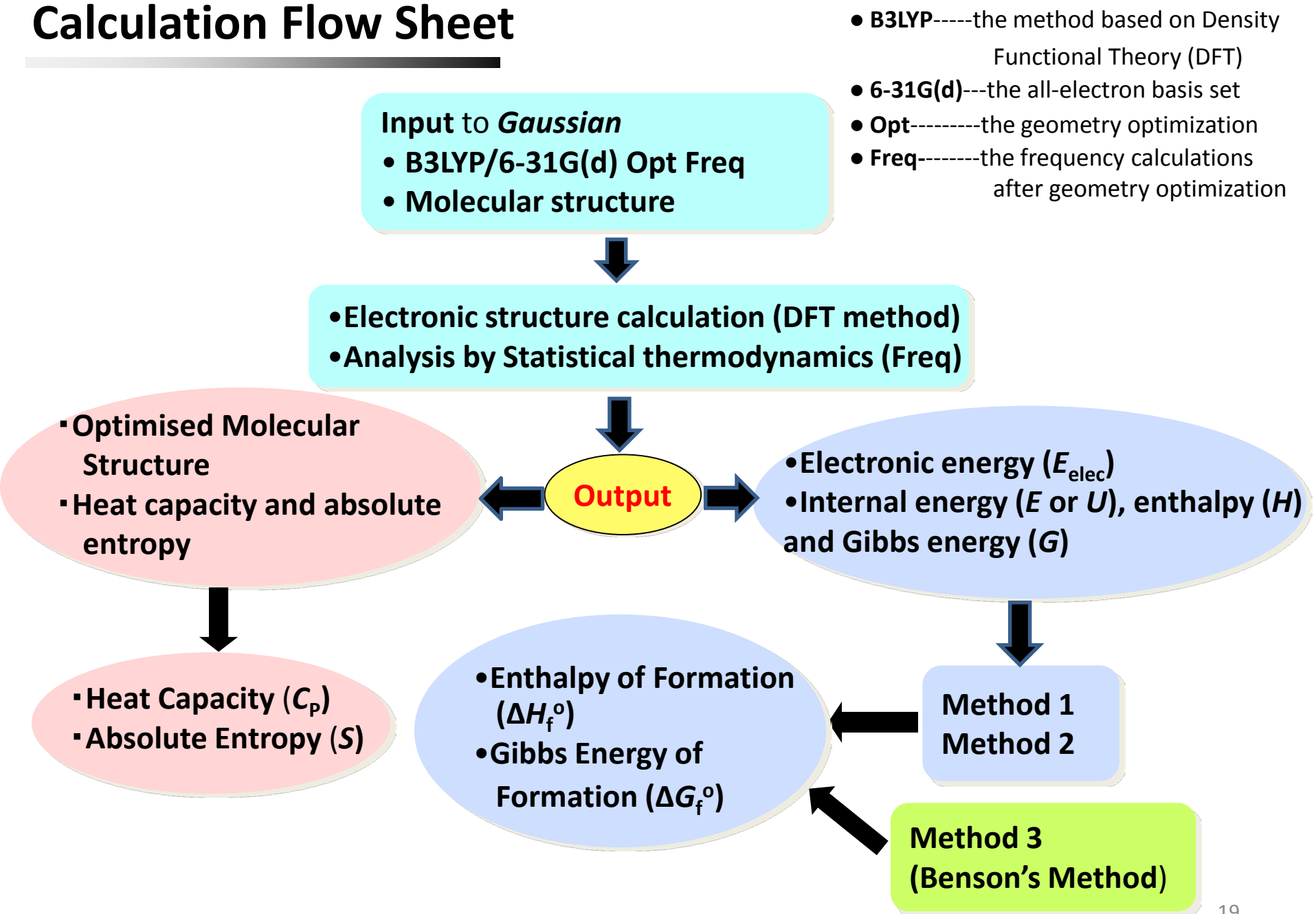
• σ - potential of the molecules, $\mu_s(\sigma)$

Second step ;

COSMO-RS (COSMOtherm)

Results : Chemical potentials of all compounds, vapour pressures, solubilities, activity coefficients.

Calculation Flow Sheet



Research subjects on Chemical Extraction of Minor Rare Metals

(1) Ionic Liquids

- Synthesis novel ionic liquids for electro-winning of MRE.
- High efficient extraction agents for the ionic liquids.
- Physical Properties measurements and simulation by first principal method
- Effect of H₂O in Ionic Liquids

Results:

- By the COSMO-RS methods, we could pick up promising ionic liquids to extract Nd and Dy from aqueous solution.
- We are now conducting extraction experiments using the picked up ionic liquids and got good results.

Developments Application Techniques

Total system development of Minor Rare Metals
using Basic researches

and

Overcome various barriers which disturb
commercialization

In Minor Rare Metals

Research subjects on Application Techniques

- (1) Total system development of Minor Rare Metals using
Basic researches
Combination a novel physical separation and extraction
techniques.**
- (2) Halogen less electro-winning media**
- (3) Clarify Bromine behaviors in PCB thermal treatments**
- (4) Pyrometallurgical processes for Minor Rare Elements
recycle from Electronics waste**

Example: New furnace for PCB

TSL Furnace : Top Submerged Lance Furnace (High performance Melting furnace)

○Merits

- Easy control of Temperature and atmosphere
- High allowance of fuels like biomass, coal, gas and oil.
- High allowance of water content in raw materials
- Easy reduction of solid waste after treatment, possible usage of molten slag and then reduce amount of harmful substances

○Raw Materials

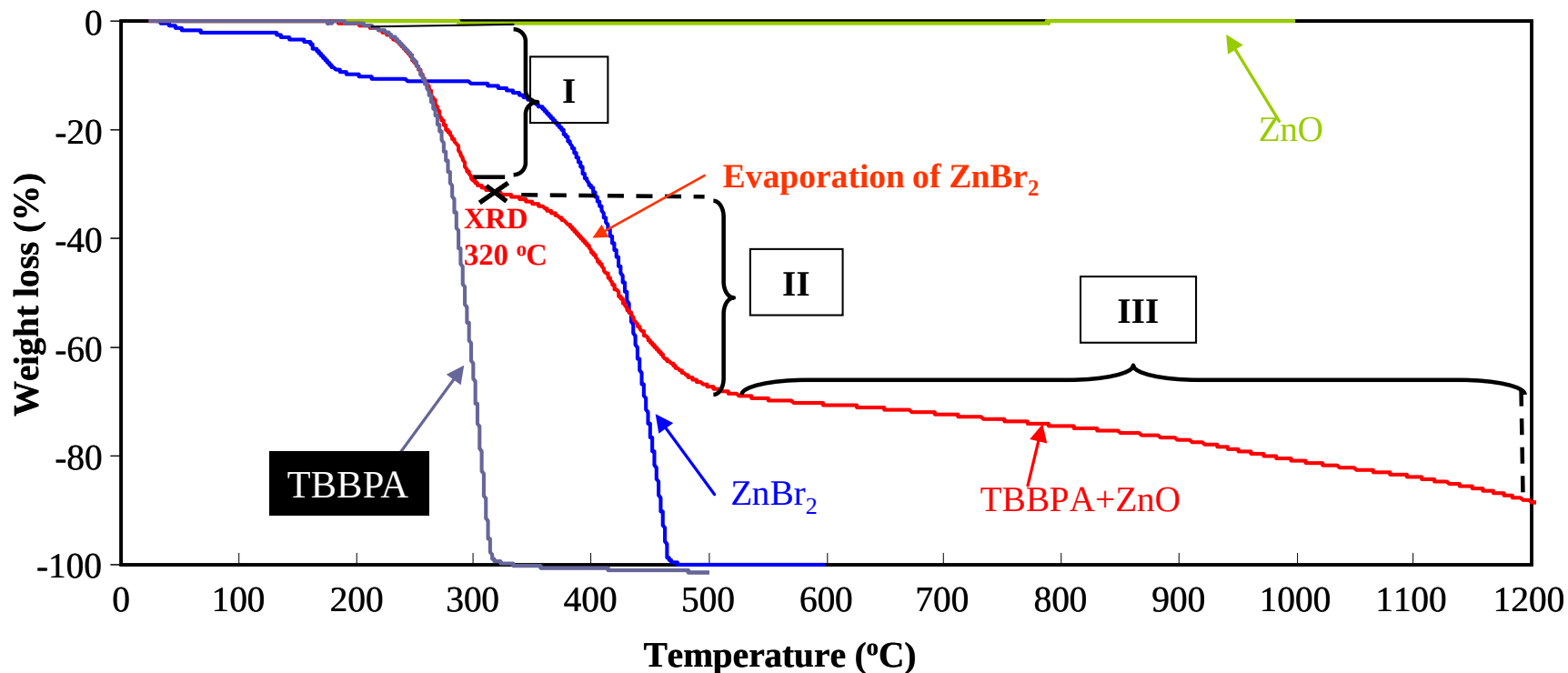
- PCB、Residue containing precious metals and other e-waste containing plastics



PCB contains Brominated flame reductants and a part of plastics also contains them. When raw materials contaminated BFRs are treated in thermal processes, bromination reactions occur and some metals like Ag, Cu and Sb are evaporated. However, their behaviors have not been clarified yet.

Basic research on Reactions between BFRs and Metal Oxides

- TG curves for mixture of **TBBPA+ZnO** (3.34:1) and TBBPA, ZnO, ZnBr₂ -



During thermal treatment of TBBPA+ZnO three stages of weight loss are observed:

TBBPA

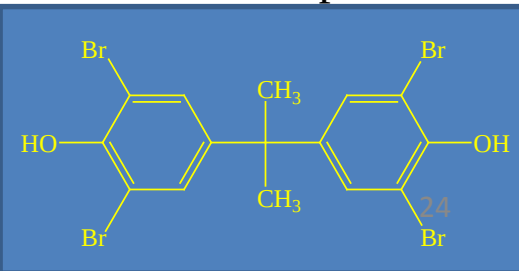
I 220-320 °C - melting and quick evaporation of TBBPA,

270-320 °C - formation of ZnBr₂

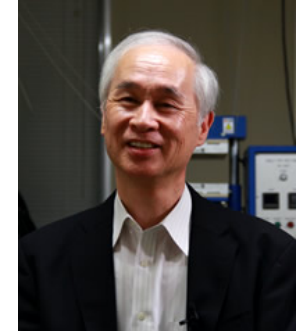
II 330-550 °C - evaporation of ZnBr₂

III 550-1200 °C - slow evaporation of organic remains

Tetrabromobisphenol A

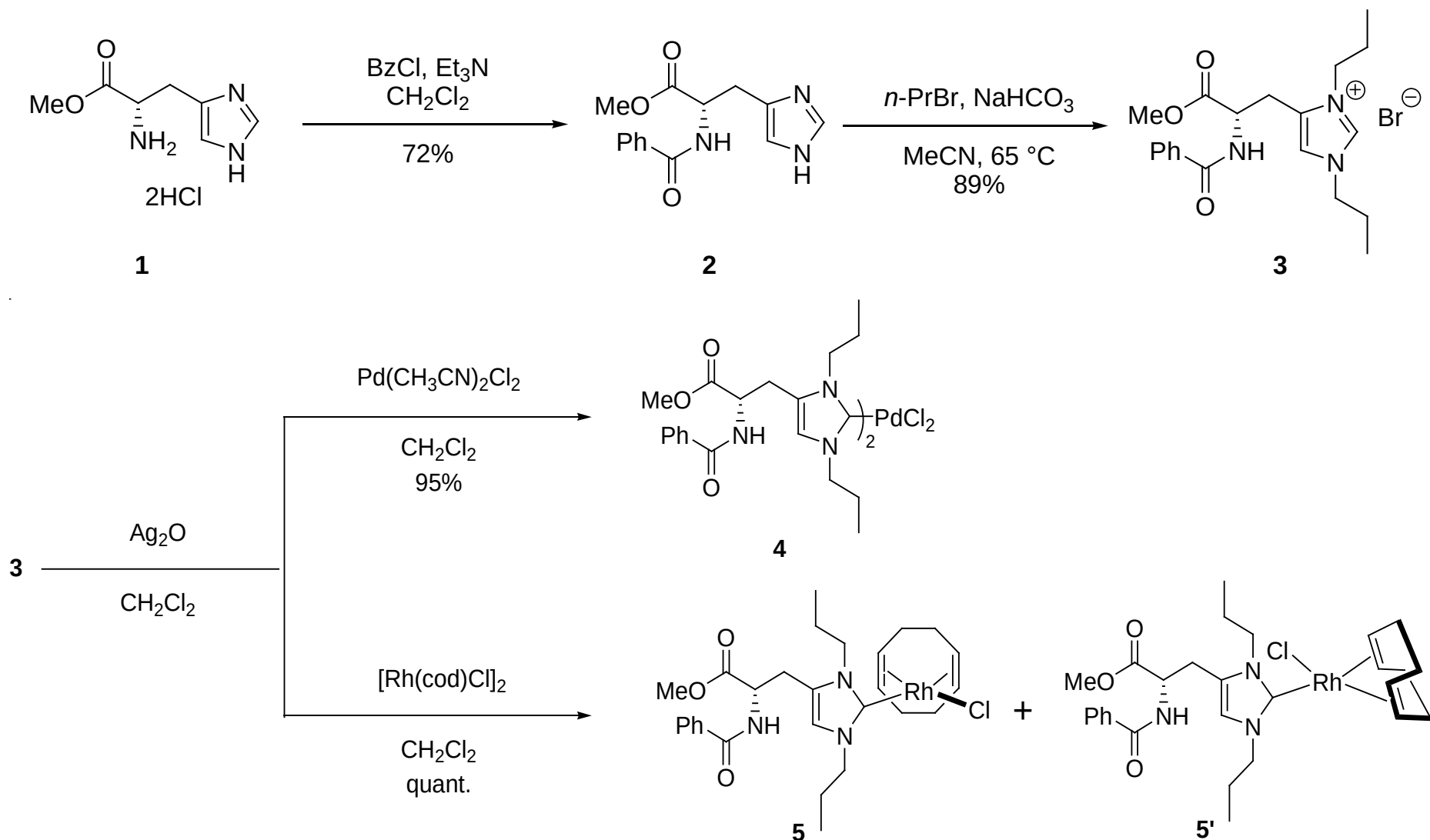


- By these researches, we would like to achieve an Industrial Innovation of MRE Recycling in Tohoku Area
- We believe we can build a new business based on the metal industry in the Tohoku region to stimulate economies in the disaster-stricken areas of March 11th earthquake and tsunami. We appreciate your support.

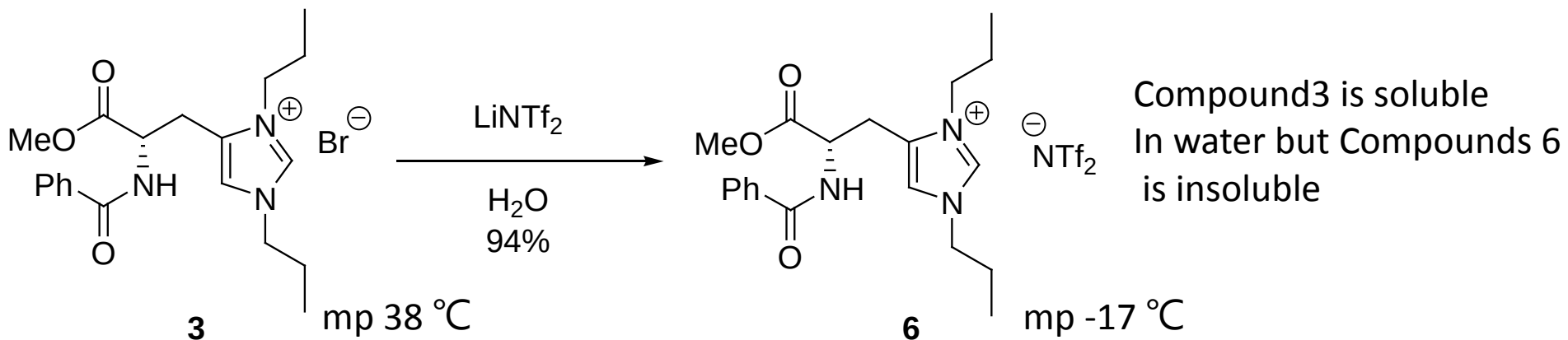


Project Leader
Prof. Takashi Nakamura

Example : Metal complex Ionic Liquid synthesis

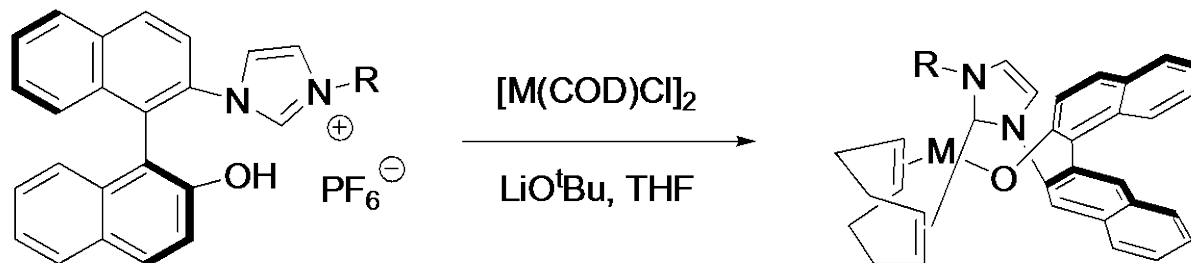
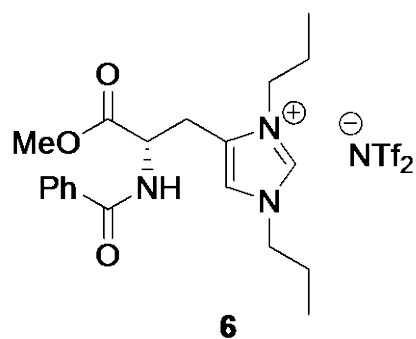


Anion exchange reactions



Next step

1. Extraction tests using compound 6
2. Design another new Ionic liquid



Behavior of Zn Evaporation in the reaction of TBPPA with ZnO

